

PRAVEEN KUMAR . E

7010335981 | Chennai - 600078 | praveen2004india@gmail.com | [LinkedIn: linkedin.com/in/praveen-kumar-e](https://www.linkedin.com/in/praveen-kumar-e)

PROFESSIONAL SUMMARY

Final-year B.Tech ECE student with hands-on experience spanning full-stack web development, deep learning, and embedded systems. Proficient in Python, PyTorch, and JavaScript with a track record of building end-to-end projects from medical imaging AI to IoT security systems. A fast learner who adapts quickly across technical domains and contributes from day one.

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML, CSS

AI / ML: PyTorch, NumPy, Diffusion Models (DDPM), OpenAI API

Web Development: RESTful APIs, Asynchronous Programming, Environment Configuration

Database: MySQL, SQLite

Embedded Systems: Arduino, Circuit Design, Hardware-Software Integration

Tools & Platforms: Git, GitHub, Google Colab, Microsoft Excel, VS Code

Soft Skills: Problem Solving, Teamwork, Communication

PROJECTS

Multimodal Brain Tumor Segmentation Using Diffusion Model | *Python, PyTorch, NumPy*

- Architected a 3D Conditional Denoising Diffusion Probabilistic Model (DDPM) in PyTorch to automate segmentation of Glioblastoma tumors from multi-modal BraTS 2020 MRI scans.
- Deployed the deep learning inference engine via a full-stack web application, enabling real-time .nii tensor processing, dynamic multi-axial color overlays, and comprehensive visualization.
- Designed end-to-end pipeline from raw MRI input to segmented output with interactive browser-based UI for medical interpretation.

AI Story Generator — AI Powered Creative Writing Tool | *HTML, CSS, JavaScript, OpenAI API*

- Built a story generation API integrating the OpenAI API to generate dynamic, context-aware stories from user prompts using asynchronous JavaScript.
- Designed RESTful endpoints with secure environment variable management; improved reliability via robust error handling and async request processing.
- Delivered a responsive front-end with real-time story streaming and a user-friendly prompt interface.

Smart Door Lock System | *Arduino, Embedded Systems*

- Engineered an Arduino-based door lock using a 4x4 keypad, servo motor, LEDs, and buzzer for secure password authentication.
- Implemented a 10-second lockout after three consecutive incorrect attempts, enhancing system security against brute-force access.
- Diagnosed and resolved Arduino IDE issues (keypad.h library conflicts, stk500_getsync() errors) through systematic hardware and software debugging.

INTERNSHIP EXPERIENCE

Embedded Systems Intern | CodeBind Technologies | Chennai, India

Jun 2025 – Jul 2025

- Assisted in the design and development of embedded systems, gaining hands-on experience with hardware-software integration.
- Contributed to system testing and debugging, ensuring robust and reliable performance across multiple embedded modules.
- Applied theoretical knowledge to practical scenarios, resolving hardware and software issues using structured problem-solving techniques.

Data Analytics Intern | SkillFied Mentor | Chennai (Remote)

May 2025 – Jun 2025

- Completed an intensive one-month Data Analytics internship; collected, cleaned, and analyzed real-world datasets using Google Colab and Microsoft Excel.

- Built data visualizations to translate complex datasets into clear, actionable business insights.
- Strengthened data accuracy and integrity by identifying and resolving dataset inconsistencies and outliers.

EDUCATION

B.Tech in Electronics and Communication Engineering (ECE)

2022 – 2026

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Chennai

CERTIFICATIONS

- Python Full Stack — SLA Institute | January 2026
- AI and ML Tools for Industrial Robots — Value Added Course | March 2024
- Web Development Internship Certificate — SkillFied Mentor | June 2025
- Embedded Systems Internship Certificate — CodeBind Technologies | July 2025